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High-fidelity multiphysics model of a permanent magnet synchronous motor for fault data generation

Hyunseung Lee[a], Seho Son[a], Dayeon Jeong[a], Kyung Ho Sun[b], Byeong Chan Jeon[b], and Ki-Yong Oh[*,a,c]

[a]*Department of Mechanical Convergence Engineering, Hanyang University, 222, Wangsimni-ro, Seondong-gu, Seoul, 04763, Republic of Korea*

[b]*Department of System Dynamics, Korea Institute of Machinery and Materials, Daejeon 34103, Republic of Korea*

[c]*School of Mechanical Engineering, Hanyang University, 222, Wangsimni-ro, Seondong-gu, Seoul, 04763, Republic of Korea*

**Corresponding author: kiyongoh@hanyang.ac.kr*

Abstract

This study proposes a high-fidelity multiphysics model of a permanent magnet synchronous motor (PMSM) for generating fault data. The proposed model helps to overcome the challenges encountered in real-world applications of data-driven prognostics and health management, where such data is often limited. To achieve this goal, i.e., generating high fidelity vibration data at a fault condition of a PMSM, the proposed model efficiently incorporates three multiphysics characteristics of a PMSM. First, a simple yet effective two-dimensional electromagnetic model is proposed to calculate electromagnetic force, which consistently induces vibrations in the PMSM. Next, the contact force within the bearing is obtained by solving analytical equations based on a rolling element bearing theory, enabling the representation of the force variation at the spall fault on the outer race. Finally, an effective modeling method for the three-dimensional PMSM structure is introduced, which is capable of generating vibration response on its surface when the Maxwell stress tensor and bearing contact force are imposed to the PMSM. The physical properties of the proposed model are estimated through extensive experiments and modal tests. The vibration data generated by the proposed model is also validated through vibration experiments, confirming the high-fidelity nature of the proposed model. The measured and generated data are also used for spall size estimation in the consideration of PMSM dynamic characteristics to demonstrate the effectiveness of the proposed model. The proposed model not only overcomes the limitation of data-driven diagnosis methods, which often lack sufficient data, but also provides a design enabling solution. As a result, the proposed high-fidelity model makes a significant contribution to the field of data-driven fault diagnosis for electric motors, offering reliable and predictive insights into bearing failures.

***Keywords**:* Bearing spall, Digital twin, Fault diagnosis, Finite element analysis, Permanent magnet synchronous motor, Vibration analysis

Nomenclature

μ	Permeability [H/m]
$\boldsymbol{J}$	Current density [A/m^2]
$\boldsymbol{A}$	Magnetic vector potential [Tm]
$\boldsymbol{B}$	Magnetic flux density [T]
$\boldsymbol{f}$	Electromagnetic force [N/m^2]
T	Torque [Nm]
L	Stack Length [m]
ω	Rotational speed [rad/s]
D_r	Diameter of the rolling element [m]

D_p	Pitch diameter [m]
K	Stiffness of the rolling element [N/m$^{2/3}$]
δ_b	Deformation of the rolling element [m]
r_r	Radius of the rolling element [m]
r_o	Radius of the outer race [m]
r_p	Pitch radius [m]
v	Poisson's ratio
E	Modulus of elasticity [Pa]
θ_{entry}	Angular position of spall entry [rad]
θ_{exit}	Angular position of spall exit [rad]
N	Number of rolling elements
ρ_0	Mass density [kg/m^3]
$\boldsymbol{u}$	Displacement [m]
$\boldsymbol{\sigma}$	First Piola-Kirchoff stress [N/m^2]
$\boldsymbol{C}$	Stiffness [Pa]
$\boldsymbol{\varepsilon}$	Strain
$\boldsymbol{f_r}$	Rotating frequency of the rotor [Hz]

1. Introduction

Recent environmental concerns have increased the use of electric motors across various industries including intelligent transportation systems and smart factories to reduce carbon emissions. This paradigm shift of industry has prompted extensive research into energy efficient motors [1] represented by permanent magnet synchronous motors (PMSMs) featuring high torque density and efficiency [2, 3]. However, motors used in diverse applications are susceptible to various adverse conditions, which in turn make them more vulnerable to failure. Especially, rotating machineries including electric motors are highly susceptible to the bearing failure due to the corrosion from external agents or excessive load variations [4, 5]. Those factors lead to the wear of races or elements of the bearings, resulting in loud noise, reduced efficiency, and even complete system shutdown, thereby causing significant economic loss or dangerous events [6]. Notably, the bearing failure accounts for 40 percent of motor breakdown, whereas other potential faults including inter-turn short circuit and misalignments exist [7]. Therefore, monitoring motor health, particularly in relation to the bearing failures, is crucial to address those issues.

Data-driven methods, enabled by advancements in data acquisition systems and sensor technologies [8], have played a significant role in the field of motor condition monitoring [9, 10]. These methods involve preprocessing vibration data and deploying sophisticated neural networks to achieve accurate fault diagnosis, covering both electrical and mechanical faults. However, data-driven methods face a significant challenge related to data scarcity in industrial applications [11]. Obtaining real-world data, especially from mechanical system fault conditions, is exceedingly challenging because such systems rarely fail naturally. Consequently, prior research on data-driven motor diagnosis had to artificially generate faults and conduct experiments [9, 10]. This approach can be cumbersome because it necessitates creating various fault modes with different extents under various operating conditions. Therefore, addressing these data scarcity issues is essential to construct efficient health management systems for real-world industrial applications.

Digital twin models have emerged as a promising solution to tackle the data scarcity challenge. These models are virtual representations of real-world systems, capable of replicating their behaviors and characteristics under identical operational conditions [12]. Researchers have recently started developing digital twin models for fault diagnosis by generating high-fidelity virtual fault data. Fault datasets at numerous conditions could be easily generated once the fidelity of the model is validated, significantly enhancing the efficiency of fault diagnosis methodology. In this purpose, a high-fidelity model of lithium-ion batteries was proposed for fault data generation to address thermal runaway issues [11]. Virtual vibration data has also been generated at progressive gear wear and spall fault at different elements in rolling element bearings (REBs) [13, 14]. The data generated through these high-fidelity models could be used to train neural networks for fault assessment, despite potential discrepancies between virtually generated and actual data. These discrepancies were mitigated through transfer learning and domain adaptation methodologies, highlighting the significant potential of virtually generated data in the fault diagnosis of actual systems. Therefore, efforts to overcome data scarcity issues by employing digital twin models need to be expanded to electric motor systems.

Finite element analysis (FEA) offers a highly accurate approach to construct high-fidelity digital twin models of motors. FEA has been widely used in noise, vibration, and harness testing and in design optimization to reduce noise and vibration through virtual simulations [15–17]. The primary source of motor vibration is electromagnetic fields, leading to in-depth study on electromagnetic fields to achieve accurate vibration response prediction. Structural responses become significant when natural frequencies of the stator are excited, necessitating sophisticated modeling that incorporates both electromagnetic and structural characteristics [18, 19]. This multiphysics characteristics could be used to identify the electrical faults with a FE model, enabling physics-based fault diagnosis [20]. However, previous models often simplified the sources of vibration for efficient goal achievements. This simplification on previous studies limits the vibration observation originating from the bearings and possible mechanical failures including bearing failure.

REBs are commonly employed in motor applications, consisting of outer race, inner race, and rolling elements. REBs smoothly interact between rotating shafts and fixed housing, generating negligible vibrations due to the balanced rotation of rolling elements (REs). However, bearings are susceptible to failure due to corrosion or excessive stress excited, leading to wear on the elements, known as spall. Spall disrupts the stable interaction between elements, resulting in excessive vibrations [21]. Specific vibration frequencies are associated with spall at each outer race, inner race, and rolling elements, enabling the diagnosis of the locations of bearing spalls. Moreover, the initiation of the spall can result in its propagation, potentially increasing the size of the spall when the faulty bearing is not managed promptly [22]. Previous research has identified that the vibration peak associated with a defective bearing comprises two distinct phenomena, corresponding to the entry and exit events of a rolling element passing over the spall on the raceway [23]. Extensive studies have been conducted to identify the points of force generation at each spall entry and exit to detect the time interval between these forces [24–27]. This time interval is crucial for estimating the size of the spall and has led to the development of various signal processing methods to maximize the detection of these events within raw vibration signals. Previous studies on bearing fault diagnosis primarily focused on dedicated bearing systems [23–27], whereas bearing fault on electric motors have not been deeply discussed. Note that the vibration signals would be weaker at the location of vibration measurements and become mixed with electromagnetic and structural characteristics in electric motors, suggesting that analyzing vibration signal in a PMSM integrated REBs is significantly challenging.

To address current gaps and limitations of data-driven fault diagnosis of PMSMs, this study proposes a high fidelity multiphysics model accounting for vibration characteristics from both electromagnetic and bearing sources in PMSMs. Specifically, this study aims to replicate dynamic characteristics of PMSMs under bearing fault conditions and thereby accurately emphasize the fault characteristics in electric motors. The novelty and key contributions of this study are as follows:

- The proposed high-fidelity model addresses the challenges related to data scarcity in current fault diagnosis methods. The multiphysics modeling method for a PMSM using FEA allows the generation of real-world replicating data, particularly under bearing fault conditions.

- The proposed model integrates electromagnetic, REB, and 3D structural models, providing dynamic responses at any location in 3D by accounting for both electromagnetic and bearing forces simultaneously. An effective 3D modeling method, coupled with extensive modal tests, ensures time-efficient yet accurate data generation.
- Two forces exciting the 3D structural model are accurately estimated from two electromagnetic and REB model. Specifically, the electromagnetic force is estimated from a simple yet effective 2D electromagnetic model using measured stator current and rotor position as input, and the contact force within the bearing is obtained by solving analytical equations based on a REB theory.
- The generated vibration data are extensively validated against measured data to confirm the fidelity of the proposed modeling method. Intuitive observation and quantitative diagnosis with appropriate processing methods on the spall fault data of a PMSM offers reliable and predictive insights into bearing failures.

This paper is organized as follows. The modeling method of a PMSM to generate the vibration responses integrating each electromagnetic, REB, and structural model is introduced in Section 2. In Section 3, the experimental setups are introduced for analysis on electromagnetic and structural characteristics and for vibration measurements under normal and bearing fault conditions. Section 4 demonstrates parameter estimation of the proposed model and comparison of measured and generated vibration data with appropriate processing for accurate spall size estimation. Finally, Section 5 concludes the paper with future work.

2. Methodology

This section presents a finite element model of a PMSM. The model incorporates electromagnetic, REB, and structural domains to generate dynamic responses of a PMSM at both healthy and bearing fault conditions (Fig. 1). First, electromagnetic model calculates electromagnetic forces with the measurements of the input current and rotor position from experiments (the blue and red box of PMSM drive input in Fig. 1), with Maxwell stress tensor method (Fig. 1(a)). Concurrently, the bearing model calculates the Hertzian contact force within the bearing with the input of the rotor position (Fig. 1(a)). These forces are then applied to the structural model simultaneously, influencing the vibration behavior of the PMSM (Fig. 1(c)). Specifically, electromagnetic force is imposed along the 3D surface of the stator core (the blue surface in Fig. 1(c)), whereas bearing contact forces for each RE are directly transmitted to the surface of the outer race attached to the casing (the red surface in Fig. 1(c)). Consequently, dynamic responses could be reliably and effectively generated. Subsequent subsections will delve into the modeling methodology for each individual domain in detail.

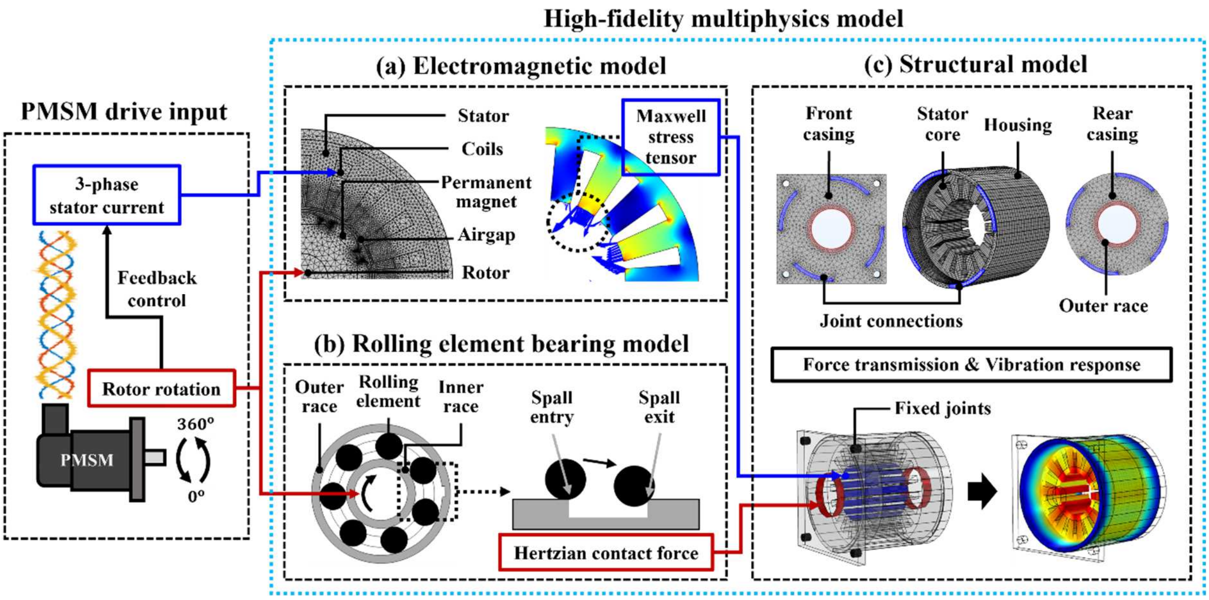

Fig. 1. A flowchart of multiphysics finite element modeling of a permanent magnet synchronous motor with (a) electromagnetic model, (b) REB model, and (c) structural model.

2.1. Electromagnetic model

The permanent magnet affixed to the rotor rotates with the excitation current in a PMSM, resulting in fluctuations in the magnetic flux within the airgap. These variations in the magnetic flux would be a main origin to generate the electromagnetic force and torque. Note that axial variations of the electromagnetic field could be disregarded because it distributes almost uniformly along the axial direction [28]. This factor suggests that modeling the cross-section of the PMSM secures sufficient accuracy on electromagnetic force and torque generated. Hence, this study addresses a 2D modeling method to calculate electromagnetic force and torque generated by the PMSM. Fig. 1(a) illustrates an axial quarter view of an 8 pole/18 slot PMSM used in this study.

The electromagnetic characteristics of a PMSM is described by Maxwell's equations as

$$\nabla \times \left(\frac{1}{\mu}\nabla \times \boldsymbol{A}\right) = \boldsymbol{J}, \tag{1}$$

$$\boldsymbol{B} = \nabla \times \boldsymbol{A}, \tag{2}$$

where μ, $\boldsymbol{J}$, $\boldsymbol{A}$, and $\boldsymbol{B}$ denote permeability, current density, magnetic vector potential, and magnetic flux density, respectively [29]. Induced magnetic flux in the air gap exerts forces on both the rotor and the stator according to the Lorentz force law when current is applied to the PMSM. This force distribution is calculated by Maxwell stress tensor method [28] as

$$\boldsymbol{f} = \frac{1}{\mu}\left[(\nabla \cdot \boldsymbol{B})\boldsymbol{B} + (\boldsymbol{B} \cdot \nabla)\boldsymbol{B} - \frac{1}{2}\boldsymbol{B}^2\right], \tag{3}$$

where $\boldsymbol{f}$ is the function of magnetic flux density, which could be mapped in the PMSM (Fig. 1(a)). Note that the blue arrows on the stator surface in Fig. 1(a) denote the electromagnetic forces, which are the main origin of the motor vibration. Indeed, the stress on the rotor surface might result in vibration; however, these forces are typically negligible because they hardly excite the quite high frequency of elastic vibration modes of the rotor [30]. Moreover, the stress distribution within the stator slot is much weaker than on the stator tooth surface. Therefore, it suffices to focus only on the electromagnetic forces acting on the tooth surface in 2D to predict the stator vibrations and dynamic responses. The magnitude of the electromagnetic force is influenced by several factors, including the material properties, and shape of each stator and permanent magnets. These parameters naturally affect the magnetic flux and consequently impact the electromagnetic force, suggesting that selecting appropriate values of electromagnetic properties plays an important role for accurate calculations of the electromagnetic forces. The parameters used in the proposed model could be found by comparing the torque value between the model and experiments in that the torque can be calculated based on magnetic flux density as

$$T = \frac{L}{\mu}\int_0^{2\pi} B_r B_\theta r^2 d\theta, \tag{4}$$

where L, μ_0, B_r, and B_θ denote the stack length of a motor, permeability of air, and radial and circumferential components of the flux density, respectively [29]. Notably, the values of parameters are not chosen arbitrarily but rather set based on the specifications and fundamental physical properties, which could satisfy the torque measurements simultaneously. These details will be discussed further in Section 4.1.

2.2. Rolling element bearing model

This subsection presents a modeling method of the vibration sources related to the REB in electric motors. The proposed method addresses analytical equations based on Hertz contact theory [32] to obtain the contact force instead of employing finite element analysis. Not only it has computational efficiency, but also numerical contact noise arises in the finite element bearing model because small impacts are periodically generated by the contact between polygonised circular elements [31]. The obtained contact forces for each RE excite the outer race, which is attached to the front and rear casing of the 3D structural model (Fig. 1(c)).

The RE rotates with the inner race affixed to the rotor, which speed depends on the bearing geometry and the speed of the rotor as

$$\omega_r = \frac{\omega_R}{2}\left(1 - \frac{D_r}{D_p}\right), \tag{5}$$

where ω_* and D_* denote the rotational speed and diameter, and subscript r, R, and p denote the RE, rotor, and pitch, respectively. The REs are compressed between the outer and inner races during the rotation due to their thermal expansion and elasto-hydrodynamic lubrication film [32, 33]. The deformation of the RE by the pressure induces contact force on the outer race, where the force could be expressed by Hertz contact theory as

$$F = K\delta_b^{1.5}, \tag{6}$$

where K and δ_b denote the stiffness and the deformation of the rolling element [32]. The stiffness K is determined by bearing geometry and material properties as

$$K = \frac{4}{3(h_r + h_o)}\left[\frac{r_r r_o}{r_r + r_o}\right]^{0.5}, \tag{7}$$

$$h = \frac{1 - v^2}{\pi E}, \tag{8}$$

where r_r, r_o, v, and E denote the radius of the RE, the radius of the outer race, poisson's ratio, and the modulus of elasticity of the bearing. The N number of contact forces would be generated around the outer race when N number of REs exist in bearing. Therefore, vibration may occur at the frequency, which is proportional to the rotational speed and number of REs. This phenomenon is commonly referred to as the ball pass frequency outer race (BPFO), which is calculated by multiplying the number of REs by the Eq. (5) [21]. Note that the value of δ_b is quite small, suggesting that vibration at BPFO from contact forces would decay in the stator housing, especially for low- to medium-power rated rotating machines [34]. Nevertheless, a substantial force variation occurs in the event of spall faults. This situation can be replicated by the proposed model, generating dynamic responses at spall faults, especially in the outer race.

Assumption made is that the contact force of a RE in the spall region becomes zero as it loses the contact with the outer race [32]. Hence, it is necessary to express this specific region by defining spall entry and exit point. In a polar coordinate, the time that RE passes the spall entry is

$$\theta_{entry} < \omega_r t, \tag{9}$$

where ω_r is angular velocity of RE and pitch radius, suggesting that $\omega_r t$ is the angular position of a RE. Describing the time when RE passes the spall exit only with angular position is insufficient because RE would change its radial position as well because of the centrifugal force, hitting the trailing edge of the spall exit (Fig. 1(b)). Here, radial acceleration by the centrifugal force is assumed constant value $r_p\omega_r^2$ because the change in radial position r_p is not big enough to change the force magnitude. The RE hits the trailing edge when distance between the position of each RE and spall exit is equal to the radius of RE r_r, so the spall region ends when

$$\sqrt{\begin{array}{l}\left(\left(\mathrm{r_p}+\frac{1}{2}\left(r_p\omega_r^2\right)(\mathrm{t}-\frac{\theta_{entry}}{\omega_\mathrm{r}})^2\right)\cos(\omega_r t)-\left(r_p+r_r\right)\cos(\theta_{exit})\right)^2\\+\left(\left(\mathrm{r_p}+\frac{1}{2}\left(r_p\omega_r^2\right)(\mathrm{t}-\frac{\theta_{entry}}{\omega_\mathrm{r}})^2\right)\sin(\omega_r t)-(r_p+r_r)\sin(\theta_{exit})\right)^2\end{array}}=r_r, \tag{10}$$

where $r_p+\frac{1}{2}\left(r_p\omega_r^2\right)(t-\frac{\theta_{entry}}{\omega_r})^2$, $\omega_r t$, r_p+r_r, and θ_{exit} represent the radial and angular position of the rolling element, radial and angular position of the spall exit, respectively. The Eqs. (5), (6), (9), and (10) are applied to all the REs by changing only the initial angular position, where they are equally positioned by $\frac{360}{N}$ degree intervals. Then, they would pass the spall region in $\frac{1}{BPFO}$ time interval. The resultant contact forces are transmitted to the outer race attached in the 3D structural model (Fig. 1(c)). Consequently, the transmission of the varying contact forces through the structural model could result in imperative vibration at spall entry and exit by a RE.

2.3. Structural model

The stator is modeled in 3D to predict dynamic responses of the PMSM at both healthy and faulty conditions. The structural parameters including geometric dimensions, material properties, and fixed constraints highly affect the modal characteristics. However, representing the entire parts of a PMSM into finite element model burdens the computation time. Hence, expressing each component in simplified ways would offer an advantage in terms of computational efficiency for reliable virtual data generation as proposed in this study.

The stator, modeled in an electromagnetic model in 2D, is extruded along the axial direction. The stator housing is connected to the front and rear casings through the bolted joints in the PMSM of interest, where casings enclose the bearing structure. Specifically, it features four bolted joints, spaced 90 degrees apart, in the front section. Similar constraints exist at the rear section. The model replicates these constraints for accurate prediction of structural responses but expressed the bolted joints into materials with high stiffness (the purple surface in Fig. 1(c)) and used the geometrically simplified casing structures. Furthermore, the PMSM would be mounted for operation, where the mounting location is in the front casings with four circular holes. Here, the elastic joints with a spring constant are used to represent this condition. Note that vibration patterns differ significantly depending on the material properties of each component and spring constants of joints, therefore, modal test was conducted to estimate the parameters, which is illustrated in Section 3.2.

The dynamic response of the elastic structure is governed by the following equation as

$$\rho_0 \frac{\partial^2 \boldsymbol{u}}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma}, \tag{11}$$

where ρ_0, $\boldsymbol{u}$, and $\boldsymbol{\sigma}$ stand for mass density, displacement, and the first Piola-Kirchhoff stress, respectively. Additional constitutive equations relate the displacement field with the elasticity of the structure as

$$\boldsymbol{\sigma} = \boldsymbol{C} : \boldsymbol{\varepsilon}\,, \tag{12}$$

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T), \tag{13}$$

where $\boldsymbol{C}$ and $\boldsymbol{\varepsilon}$ are the stiffness and the strain tensors [18]. The configurations of the forces on its structure are shown in Fig. 1. Specifically, the electromagnetic force calculated in the 2D electromagnetic domain with input current and rotor's rotational position is imposed along the 3D surface of the stator core (the blue surface in Fig. 1(c)). The Hertzian contact forces between outer race and each rolling element, determined by the rotor's rotational position and the spall entry and exit conditions, were also directly transmitted to the outer race attached to the casing (the red surface in Fig. 1(c)).

2.4. Construction of the model

In this study, the finite element model was constructed using COMSOL Multiphysics (COMSOL Incorporated, USA). Both 2D electromagnetic and 3D structural models were established, with specific geometric and material properties detailed in Section 4, along with experimental validation. The electromagnetic model near the air gap was finely meshed to enable accurate electromagnetic force calculations on the stator teeth surface (Fig. 1(a)), resulting in total 24,973 mesh elements. Similarly, 73,718 mesh elements were used for the structural model (Fig. 1(c)), and the circular path of the outer race, where contact forces from REs rotate around, was finely meshed. Fine meshing was also applied to the stator structure, given its significant influence on the modal characteristics. The parameters for the bearing model were derived from the actual measurements of bearing, introduced in Section 3.4. Each electromagnetic model and bearing model calculate forces based on input current and rotor position, as illustrated in Fig. 1. These forces influence the structural response of the constructed 3D model. Consequently, the structural vibration could be obtained at any location within the 3D model, facilitating the observation of varying responses based on the bearing health status.

3. Experiments

This section introduces a series of experiments conducted to estimate parameters and validate the vibration responses of the proposed model. The torque measurements and modal tests were necessary to determine the material properties of the electromagnetic and structural models, where the bearing model was assumed to have steel properties at room temperature. The dimensions of the model could be determined based on the measurement of the PMSM of interest. Experimental validation for vibration responses was conducted by acceleration measurements under both bearing healthy and faulty conditions, where different widths of spall faults were artificially seeded on the outer race of the bearing.

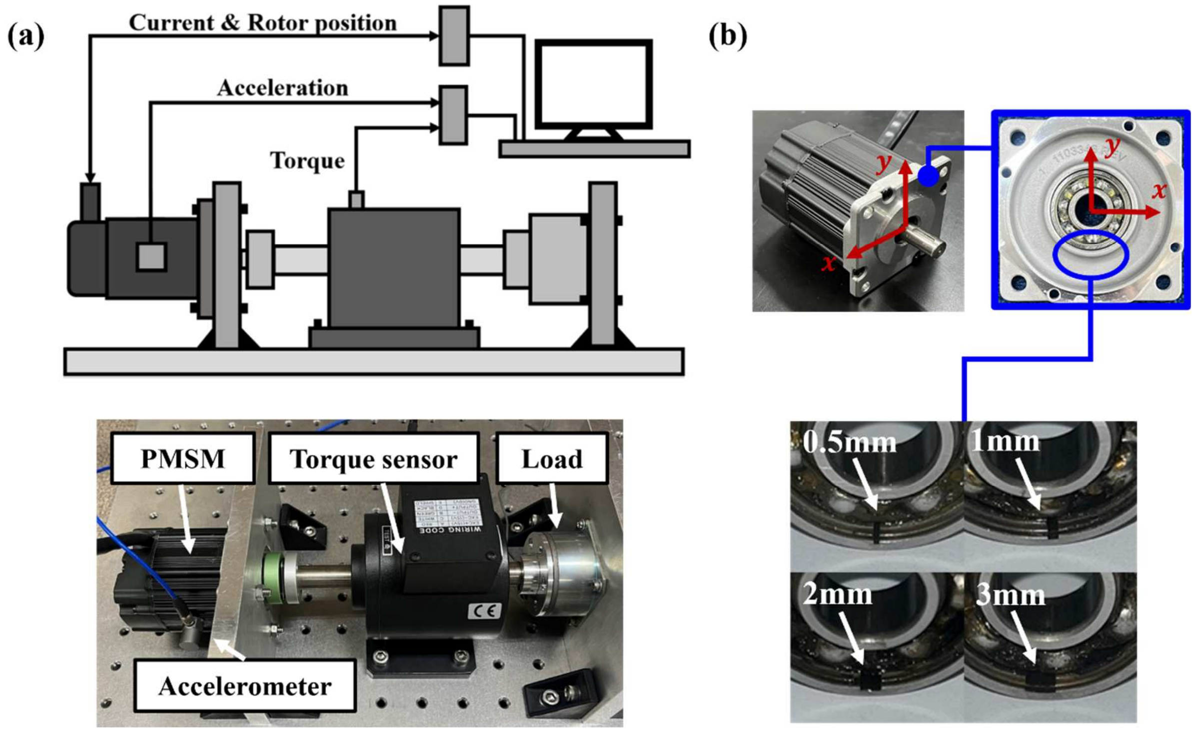

Fig. 2. (a) PMSM testbench construction for data acquisition, (b) PMSM bearing configuration and machined outer race spall fault at different widths.

3.1. Testbench construction and setup

The testbench was set up with an 8-pole/18-slot surface-mounted PMSM (M-2310P, TEKNIC, USA) (Fig. 2(a)). This brushless rotary motor features high torque density with continuous rated torque of 0.28 Nm and rated speed of 6,000 RPM. The NSK 6000 bearings were also installed on both front and rear casings of the PMSM (Fig. 2(b)). Data acquisition was performed using a driver (DRV8305, Texas Instruments, USA), a torque sensor (TMRA-0.5K, Setech, Republic of Korea), and an accelerometer (352c33, PCB, USA). Specifically, the PMSM was operated using field-oriented control (FOC) technique, facilitated by the built-in encoder, enabling precise rotor position measurement and speed feedback. Hence, the current and the rotor position were obtained by the driver during the operation. The torquemeter was connected to the PMSM rotor for the purpose of quantifying torque value under several operational conditions to determine and validate parameters of the electromagnetic model. A uniaxial accelerometer was also installed on the stator housing to monitor the radial vibrations, i.e., the primary dynamic behaviors of the PMSM. The torque and acceleration sensors were connected to the DAQ (NI 9205 and NI 9234, respectively, National Instruments, USA) for data acquisition.

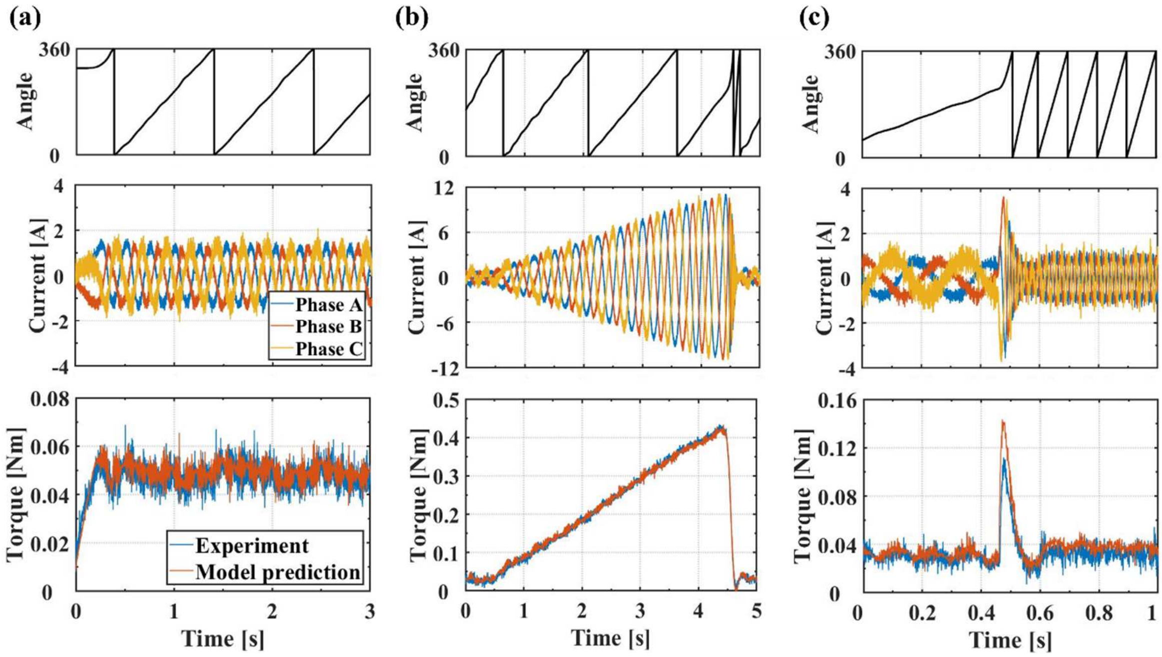

Fig. 3. Torque validation of the electromagnetic model with measured current and rotor position at each operational condition (a) steady-state condition, (b) load varying condition, and (c) speed varying condition.

3.2. Torque measurements

The torque was measured from the torque sensor with a healthy PMSM under several operational conditions. This experiment aims to not only estimate parameters of the electromagnetic model but also validate the electromagnetic responses under different operating conditions. The experiment aimed to align the torque responses with those predicted by the electromagnetic model, utilizing input physical parameters including 3-phase current and rotor position. Therefore, a series of experiments were conducted, where a steady-state experiment was performed to fine-tune the parameters of the electromagnetic model, whereas transient response experiments were conducted to verify that the fine-tuned electromagnetic model could accurately predict the torque responses under varying input datasets. Varying data sets of such parameters were accumulated through following experiments. First, the PMSM was tested from non-operation to 60 RPM without specific load applied. The torque was measured for three seconds, and the steady-state electromagnetic response was compared under corresponding condition (Fig. 3(a)). The second operational condition was performed by progressively increasing the load from free to the maximum, maintaining a constant commanded rotor speed (Fig. 3(b)). The current would increase to generate greater torque when external load is applied, thereby maintaining the specified speed. This approach enables meaningful torque comparisons across a wide current range in a wide torque region of interest. The external load was applied to the extent where the torque slightly exceeds the continuous rated torque of 0.28 Nm, with a speed command of 60 RPM. The test duration was set to five seconds. The last experiment varied the speed command from 60 RPM to 600 RPM, in a very short period, with a free external load (Fig. 3(c)). The current demand would increase to bridge the speed differential between the reference and feedback speeds, similar to the prior operational scenario. This test was conducted in one second, with the speed command transitioning directly from 60 RPM to 600 RPM in a single step, enabling the observation of the immediate electromagnetic response of the PMSM. The torque value was compared with the variation of the external conditions to enhance the reliability of the estimated parameters. Note that the current and rotor position measured from experiments were identically fed into the electromagnetic model to calculate the torque.

3.3. Modal testing

The modal test was conducted to investigate the structural characteristics of the PMSM. This experiment aims to characterize equivalent structural properties of the PMSM, which would be used for the proposed modeling approach. Specifically, the PMSM mainly comprises stator core, housing, and casings (Fig. 1(c)). Usually, the laminated steel, aluminum, and steel are employed for corresponding assemblies in PMSM, which was similar in the tested PMSM. However, selecting the exact properties of each material for the structural model may result in considerable difference in modal characteristics because the model could not represent the geometric dimensions and material properties of real-world structures perfectly. Even more, the proposed model does not account for the coil structure and stator skew shape for effective data generation purpose. The front and rear casings were also geometrically simplified, where bolted joints are represented in elastic materials, suggesting that finding the equivalent stiffness of the connecting material between the stator housing and casings is simple yet effective way to accurately estimate structural dynamics of the PMSM. All the necessary parameters could be determined by comparing the modal test results and mode analysis of the structural model.

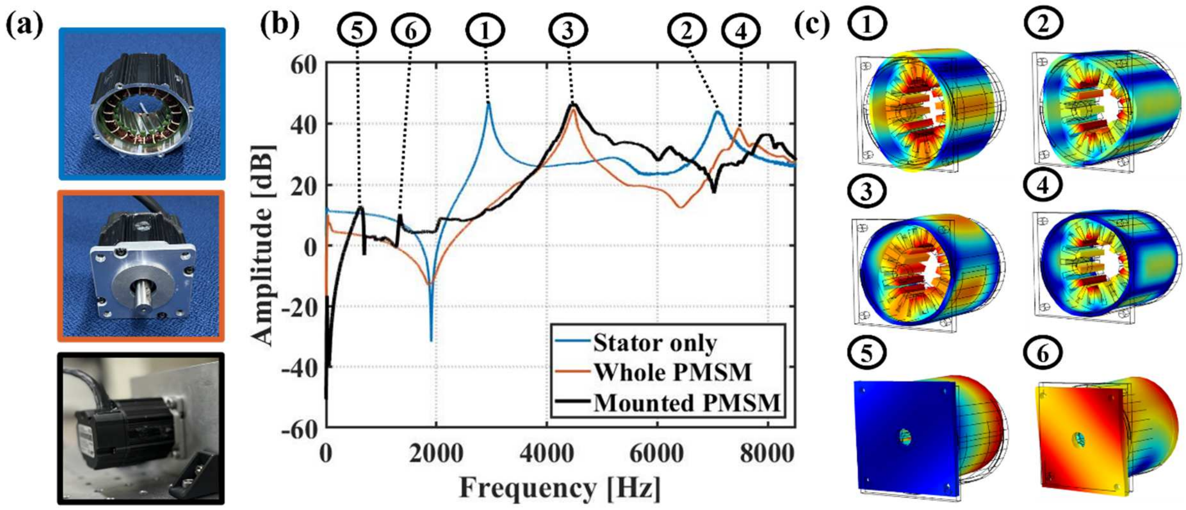

Fig. 4. (a) PMSM at each joint condition for modal tests, (b) experimental frequency response function at each joint condition, and (c) calculated mode shapes at corresponding conditions with FE structural model.

The impact hammer (086c03, PCB, USA) and accelerometer (352c33, PCB, USA) were used to measure the free response at impulse experiments. The hammer was equipped with a steel tip to induce an impact force on the stator housing. Here, the PMSM was dismounted from the testbench and positioned on a soft material to isolate it from the external vibrations. The accelerometer was attached to a point of a stator housing, where vibration was measured for radial direction. Note that the electric motor stators typically exhibit radial vibration modes, such as elliptical, triangular, and rectangular shapes [35, 36]. The frequency response function could be calculated with measured force and acceleration with PULSE 3050-A-060 (B&K, Denmark), identifying the natural frequencies. The mode shapes at each natural frequency were not obtained experimentally, however, they could be estimated with the shapes calculated by the FE model at the corresponding frequency.

The PMSM was tested on several conditions to identify and distinguish the effect of material properties and joint conditions on modal characteristics (Fig. 4(a)). First, only the stator without the front and rear casings were tested to determine the material properties of the stator because the bolted joints on the housing would vary the modal characteristics of the stator. Here, the equivalent parameters of the stator were determined by comparing the natural frequencies with those of a structural model without the casing connections (the blue line in Fig. 4(b)). Second, the entire PMSM structure was tested to estimate the properties of connecting materials and the casings (the orange line in Fig. 4(b)). Third, the mounting spring constants were determined through the modal test (the black line in Fig. 4(b)), where the PMSM was secured onto the testbench. This experiment allowed the identification of the vibration modes due to the relation between the PMSM structure and the mounting conditions, whereas first and second tests identify the modes originated from the elasticity of the stator core, housing, and casings. The detailed analysis on the comparison of the natural frequencies based on estimated structural parameters are discussed in Section 4.2.

3.4. Fault generation and vibration measurements

The NSK 6000 bearings were installed on the PMSM of interest (Fig. 2(b)). Geometric parameters were measured for the bearing model (Table 1). These bearings were machined to induce spall faults on the outer race by employing an electro machining method [21], which allows accurate machining in geometry and size of the spall. The spall faults with depth of 0.5 mm and width of 0.5-, 1-, 2-, and 3-mm on the outer race were seeded by first making a fine hole through axial direction with a super drill followed by wire cutting (Fig. 2(b)). Subsequently, the bearing was assembled onto the PMSM with the spall positioned approximately positive 4.7 radians from the x-axis shown in Fig. 2(b). Total four bearing fault conditions were considered in this experiment.

The PMSM vibration was measured on the testbench with the accelerometer. PMSM with healthy bearing was first tested in wide operational speed range from 1,000 to 5,000 RPM at 50 RPM intervals, which facilitate the identification of vibration sources across the broad frequency range. Then, the vibration spectrums of the experiment and the model were compared at the representative speed of 3,000 RPM, which is the half-rated speed of the PMSM. The vibration was measured for 0.5 seconds, which was considered sufficient time to observe the steady-state vibration of the healthy PMSM. It was sampled at 25,600 Hz; therefore, the frequency resolution would be 2 Hz. The PMSM under four different bearing fault conditions were also tested on the testbench. They were operated at 1,000 RPM, and transient responses were observed. The validation of dynamic responses with appropriate signal processing will be discussed in Section 4.3 and 4.4.

Table 1 Geometric parameters of the bearing model.

Specification	Value
Number of rolling elements	7
Diameter of outer / inner race	26 / 10 mm
Pitch diameter	18 mm
Diameter of RE	4.5 mm
Deformation of RE	0.01 mm
Location of spall	4.7 rad
Spall width	0.5, 1, 2, 3 mm
BPFO value	$2.625 \times f_r$ Hz

4. Results and discussion

4.1. Parameter estimation and validation for an electromagnetic model

Electromagnetic parameters were estimated through three steps. First, the cross-sectional geometry of the PMSM was measured. The measurement included the air gap length, stator and rotor diameters, stack length, coil winding, and the shape of slot opening and permanent magnet with the PMSM (M-2310P, TEKNIC, USA) (Table 2). Second, materials for each component were determined, where copper was used for coil, soft iron for the stator and rotor, and neodymium for the permanent magnet [38, 40]. Finally, some of the parameters, which have great influence on torque responses, were fine-tuned. Only a few critical parameters were adjusted to simplify and streamline the parameter estimation process. These materials include magnet remanence, slot opening, and a length of major/minor axis of permanent magnet which directly affect the torque magnitude and cogging torque behavior, respectively [41]. Note that the differences in geometry measurements and stator skew, which is not considered in the cross-sectional model might result in the error without the equivalent parameters aforementioned when predicting the electromagnetic responses. Hence, parameter estimation was executed with measurements during steady-state operation (Fig. 3(a)). The accuracy of the model with estimated parameters was demonstrated using the root mean squared error (RMSE), as shown in Table 3. It was calculated by subtracting the torque magnitudes of the experiment and model at each time point within the time period of experiments. Moreover, the torque graph visually illustrates that the model predicted values closely matched the actual response with adjusted magnet remanence as shown in Table 2. Notably, the remanence value falls within the reasonable range for commercially used permanent magnets [39], where experimental PMSM uses the sintered Neodymium magnets. Furthermore, the fluctuations in the electromagnetic response were observed due to cogging torque (Fig. 3(a)), primarily arising from the interaction between the stator slot and permanent magnet shape. The geometric values of these components were determined based on the cross-sectional view of an experimental PMSM. Magnet lengths were estimated based on its elliptical shape. The RMSE value was also low with slight error possibly attributable to noise in the measured data (Table 3). This visualized and quantitative comparison confirms the reliability of the generated electromagnetic response based on the estimated parameters.

Table 2 Parameters used in the electromagnetic model.

Specification	Value	Origin of value used
Airgap length	1.0 mm	Measured
Stator diameter	50.0 mm	Measured
Rotor diameter	22.0 mm	Measured
Stack length	23.0 mm	Measured
Coil winding	Distributed	Measured
Electrical conductivity of stator and rotor	0 S/m	[40]
Relative permittivity of stator and rotor	1	[40]
Electrical conductivity of coil	5.998×10^7 S/m	[38]
Relative permittivity of coil	1	[38]
Magnet remanence	1.47 T	Estimated
Slot opening	1.4 mm	Estimated
Length of major / minor axis of permanent magnet	7.0 / 3.0 mm	Estimated

Two experiments were also conducted under variable operational conditions to validate the accuracy of the proposed electromagnetic model. Fig. 3(b) shows the comparison under load varying condition. The current increases with the external load in this experiment, allowing for the observation of the torque response across the entire current range. It can be observed that the model response closely tracked the actual response with the estimated parameters. The torque value is nearly proportional to the input current, as evident in both the experiment and model. The RMSE values remained low, similar to those observed during the steady state comparisons. It is important to note that the error scale remained consistent although the overall torque magnitude is higher in this condition, suggesting that the error likely stemmed from the same source, i.e., noise in the measurement data. This comparison under variable current input underscores the robustness of the electromagnetic model, by demonstrating that the model successfully predicts the torque values under the current which transiently changes in a wide operating range. Next, Fig. 3(c) shows the comparison under speed varying condition where the command speed was abruptly altered at 0.5 seconds. This change is clearly visible through the increased frequency of measured current and rotor position at a given time. Moreover, the current magnitude immediately rises, leading to a sudden torque increase in a very short time. The proposed model well account for this phenomenon with the inputs of the measured current and rotor position. However, the RMSE value for this condition was higher than those in other conditions. The discrepancy in the maximum peak torque would be an origin for this high RMSE. This difference may originate from the model's inability to fully reflect the steep change in driving input or from errors in the data collected under abnormal and harsh operating condition, changing command speed of 540 RPM at once. Nevertheless, reducing this difference was beyond the scope of this study in that it was an unusual and extreme operation, and the vibration prediction for the PMSM was conducted under the constant speed command. Still, it is noteworthy that the proposed model could even capture the sudden change in the electromagnetic responses.

A series of experiments confirm the accuracy of the predicted torque under steady current and rotor position conditions and demonstrate that the proposed model well replicate the electromagnetic responses under extreme conditions such as rated load and rapid speed change. This comprehensive validation underscores the reliability and robustness of the proposed electromagnetic model with given parameters, suggesting that magnetic flux can be accurately calculated in a variety of operating conditions. Consequently, the electromagnetic force, significant source of vibration in a PMSM, can be precisely obtained, aligning with real-world observations.

Table 3 RMSE between the experimental measurement and model prediction

Condition	Compared time range [s]	RMSE [Nm]
Steady state (Fig. 3(a))	0.5 – 3.0	0.006
Load variation (Fig. 3(b))	0.5 – 4.5	0.007
Speed variation (Fig. 3(c))	0.4 – 0.6	0.018

4.2. Parameter estimation and validation of a structural model

The dimensions of the stator were measured for constructing the 3D stator model, whereas the front and rear casings of the PMSM were simplified as square and circular shapes, respectively (Fig. 1(c)). Notably, the rotor was omitted from the structural model because its natural frequency and mechanical force have minimal impact on the vibration of the stator surface [30, 34]. The materials used for the stator core and housing were made from laminated steel and aluminum, and front and rear casings were made from steel and aluminum. However, the exact material properties were fine-tuned until the structural model could accurately replicate modal characteristics of the actual PMSM because the model with initial parameters would result in different characteristics. Specifically, the structural model did not account for the stator skew shape and coil windings, which could influence the natural frequencies. These parameters could alter the mass and overall stiffness of the stator structure [42], necessitating the estimation of equivalent parameters when they are not included in the structural model for simplification. Additionally, stator core for an electric machine is usually multi-layered orthotropic laminates [43], whereas the proposed structural model assumes isotropy. Hence, modal tests were conducted to measure frequency response functions at each condition that played an important role in estimating several parameters related to structural dynamics of the PMSM (Fig. 4).

Table 4 Parameters used in the structural model.

Component	Specification	Value	Origin of value used
Stator (core / housing)	Axial length	23.0 / 46.0 mm	Measured
	Young's Modulus	158 / 63 GPa	Estimated

	Density	7,600 / 2,600 kg/m^3	Estimated
	Poisson's ratio	0.3	Estimated
Casing (front / rear)	Young's Modulus	170 / 50 GPa	Estimated
	Density	7,600 / 2,800 kg/m^3	Estimated
	Poisson's ratio	0.3	Estimated
	Young's Modulus of simplified bolted joints	120 GPa	Estimated
Mounting joint	Spring constant	2.2e6 N/m	Estimated

Two natural frequencies were identified when testing only the stator structure, which includes the core and housing. The first and second mode shapes of the stator are typically elliptical (2nd) and triangular (3rd), stemming from the elasticity of the stator [35]. Same mode shapes were obtained through the structural model when the casings were not included (two mode shapes at the top of Fig. 4(c)). It is reasonable to assume that the experimental mode shapes are similar to these because they are major mode shapes of the hollow cylindrical structure, and the natural frequencies at each shape were similar without any fine-tuning of the properties. Still, their value could not be the same due to the discrepancies in geometry and material properties. Equivalent material properties of the stator core and housing were determined to minimize this difference in natural frequencies. The obtained parameters and natural frequency comparison are listed in the first part of Table 4 and Table 5, respectively. These parameters, excluding dimensions, were fine-tuned not only to match the natural frequency characteristics but also to maintain values consistent with typical PMSM attributes. It is worth noting that the Young's modulus of laminated steel, commonly used in stator cores, is much lower than that of regular steel [42], so the listed material properties are aligned with those of PMSMs.

Natural frequencies increased when the front and rear casings were attached with bolted joints. In particular, the natural frequency of the second mode shape increased by 152 %. The casings provided resistance to deformation from original mode shapes, resulting in an overall increase in structural stiffness and higher natural frequencies. Then, the mode shape slightly changes especially near the stiffened regions (two mode shapes at the middle part of Fig. 4(c)). The configurations of casings and the bolted joints (which were represented by a single elastic material) led to different modal characteristics, and equivalent properties for them were estimated in this experiment. Estimated parameters and natural frequency comparison are shown in the second part of Table 4 and Table 5, respectively. The increase in natural frequencies was accurately predicted within approximately 3% error, suggesting that representing the bolt conditions in such a way could be simple yet effective for replicating structural characteristics of the stator. Otherwise, the increase in natural frequencies with that amount for each mode could hardly be explained. Therefore, a simplified four-featured joint condition was necessary to strike a balance between accuracy and computational efficiency.

The final modal test result with the mounted PMSM showed similar natural frequencies regarding stator modes because no specific joints were added into the stator housing and core. However, new distinct natural frequencies were observed in low-frequency region at 634 and 1,341 Hz. The new constraints introduced the relative motion between the PMSM and mounting joints, referred to as the first and second mounting modes in this paper. Similar fixed joint conditions were applied in the 3D structural model to account for this phenomena, and new mode shapes appeared in a similar frequency region (bottom two mode shapes at Fig. 4(c)). The spring constant for the fixed joint conditions was tuned, and natural frequencies for a total four modes were compared (third part of Table 5). The mounting modes are not typically expected to appear in healthy motor vibrations because the net force in the translational direction hardly appears when electromagnetic force is evenly balanced along circular airgap. However, a net force can be generated in any directions when faults occur, resulting in a translational force, so these vibration modes could occur significantly. The proposed model could not only replicate vibration modes due to stator elasticity but also account for the mounting modes thanks to its consideration of the entire PMSM operating conditions.

The proposed structural model could replicate the structural characteristics with subsequent modal test results. Appropriate modal tests were conducted considering the way of modeling, so that equivalent parameters were found component by component which enabled the simplification of the model successfully. As a result, the structural model efficiently reflected the modal characteristics of the PMSM, as validated through several natural frequency comparisons. Furthermore, the determined parameters are aligned with the actual material properties, underscoring the reliability and robustness of the proposed structural model.

Table 5 Comparison of natural frequencies of the experiment and model.

Condition	Vibration mode	Experiment	Model	Error
Stator only	2nd stator mode	2,950 Hz	2,857 Hz	-3.2 %
	3rd stator mode	7,093 Hz	7,095 Hz	0.3 %
Whole PMSM	2nd stator mode	4,484 Hz	4,434 Hz	-1.1 %
	3rd stator mode	7,467 Hz	7,706 Hz	3.2%
Mounted PMSM	1st mounting mode	634 Hz	631 Hz	-0.5 %
	2nd mounting mode	1,341 Hz	1,454 Hz	8.4 %
	2nd stator mode	4,493 Hz	4,451 Hz	-1.0 %
	3rd stator mode	7,943 Hz	7,713 Hz	-2.9 %

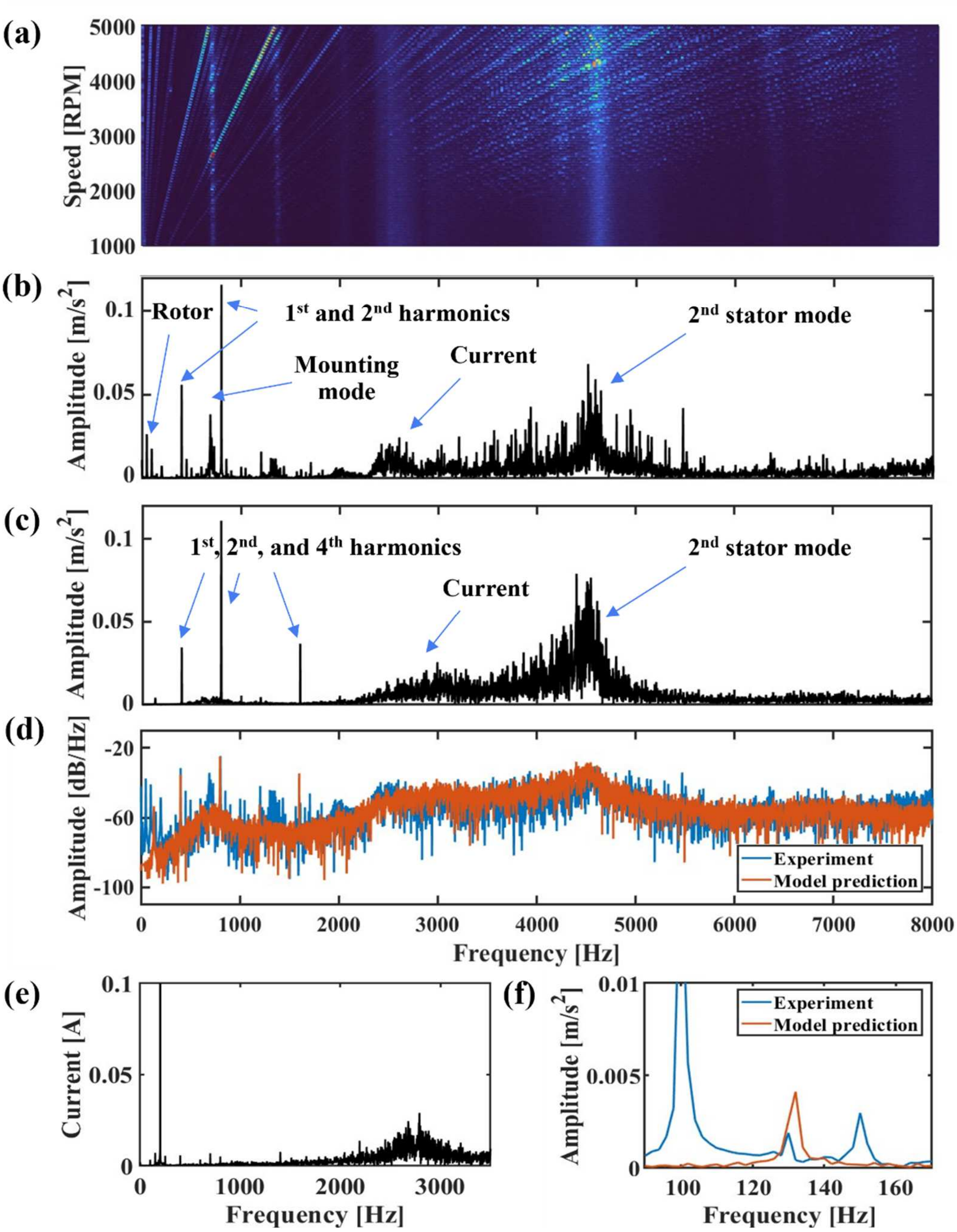

Fig. 5. (a) Waterfall plot of the experimental vibration, (b) experimental vibration in frequency domain at 3,000 RPM, (c) model predicted vibration in frequency domain, (d) power spectral density comparison between the experiment and the model prediction, (e) measured current in frequency domain, and (f) BPFO comparison between the experiment and model prediction.

4.3. Vibration response at normal operational conditions

The vibration response from the proposed model was validated with the experimental measurements from a PMSM with a healthy bearing. Specifically, the responses of the PMSM were compared during steady-state rotation at 3,000 RPM with no external load. In this comparison, the model's acceleration was calculated in the time domain over a 0.5-second duration and sampled at the frequency of 25,600 Hz at the same location, i.e., midpoint of the stator in the radial direction, to match the frequency resolution of the experiments for fair comparison. The comparison was facilitated by converting the time domain data into the frequency domain through the Fast Fourier Transform (FFT), which is particularly effective for analyzing steady-state vibrations. The waterfall plot was presented to aid in identifying the vibration sources in the frequency domain (Fig. 5(a)), and the acceleration was compared (Fig. 5(b) and (c)). The frequency components were clearly distinguished in the waterfall plot, some being proportional to the rotating speed and others not. The structural response has four characteristics, which could be originated from electromagnetic and bearing contact forces.

First, the primary source of vibration under a normal operational condition is electromagnetic force from the rotation of permanent magnet and the flow of stator current. This primary source could result in vibration at eight times the rotating frequency because of an eight-pole of the PMSM. Hence, the corresponding frequency, i.e., 400 Hz, and its harmonic components were observed in both the experiment and model. Specifically, the model exhibited dominant first, second, and fourth harmonics (400, 800, and 1600 Hz), whereas only the first and second harmonics (400 and 800 Hz) are distinct in the experiment. This difference would be attributed to the slight mismatch in the vibration measurement locations. The vibration at different points on the stator housing could vary because some points are directly above the stator teeth where electromagnetic force is distributed, whereas others are not. Furthermore, the presence of stator skew typically reduces the higher harmonics [20]. The model managed to replicate the same frequencies with similar magnitudes of the major harmonic components.

Second, the electromagnetic force is also influenced by the current in that the measured current had frequency components near 2,500 Hz in this experiment (Fig. 5(e)). Note that excitation from current input contributed to the vibration in both the experiment and model responses. This analysis clearly validates that the electromagnetic parameters estimated is accurate. The direct use of experimental current and rotor position also enables an accurate replication of the vibration caused by electromagnetic force with the proposed model.

Third, the structural responses, vibrating at their natural frequencies, consistently exhibited vibration during steady-state operation. In particular, the second stator mode near 4,500 Hz was dominantly excited across the entire speed range (Fig. 5(a)). The proposed model closely tracked the excitation of the actual system by replicating modal characteristics with validated structural parameters. However, a noticeable discrepancy in vibration at mounting mode frequency was evident at 705 Hz in the spectrum (Fig. 5(b) and (c)). The experiment showed an amplitude of nearly 0.05 $\mathrm{m/s^2}$ at the corresponding natural frequency, whereas it was hardly predicted by the model. This difference could primarily be attributed to the absence of rotor's response in the model. Specifically, rotors have the potential to generate periodic translational forces due to inherent mechanical imbalances, including unbalance mass and misalignments. These mechanical forces could excite the mounting mode during the experiment but are not accounted for in the model. Vibration from these forces was also observed only in the experiment at the rotating frequency of 50 Hz and its harmonics, which is a common occurrence in most rotating machinery (Fig. 5(b)). However, this vibration source was not significant across the entire frequency range and could be disregarded. Excluding the vibration generated by the rotor, the proposed model replicated a similar response in combined electromagnetic and structural characteristics across the entire frequency range. This similarity is also evident in the power spectral density plot depicted in Fig. 5(d), which compares the experimental and model-predicted vibrations, demonstrating the similarity in vibration caused by electromagnetic force and structural response at natural frequency.

Fourth, vibration caused by the presence of the bearing was hardly visible when bearing was in a healthy condition. Therefore, the vibration frequency caused by the bearing contact force was magnified to make it visible for comparison (Fig. 5(f)). The characteristic frequency of BPFO could be obtained from the bearing specifications in Table 1, which is 131.25 Hz, suggesting that the model closely matched the vibration at this frequency. Note that the frequency components at 100 and 150 Hz in the experiment were harmonics of the rotating frequency. The magnitude of BPFO in the experiment was only half as much as that in the model, because BPFO dissipates more easily at the measurement point when passing the boundary between the outer race and the casing and passing the joints between the casing and housing. However, this difference in BPFO magnitude did not significantly impact the overall vibration characteristics under normal operational conditions. The important point is that the vibration due to the bearing contact force could be successfully replicated from the combination of the bearing model and simplified structural model.

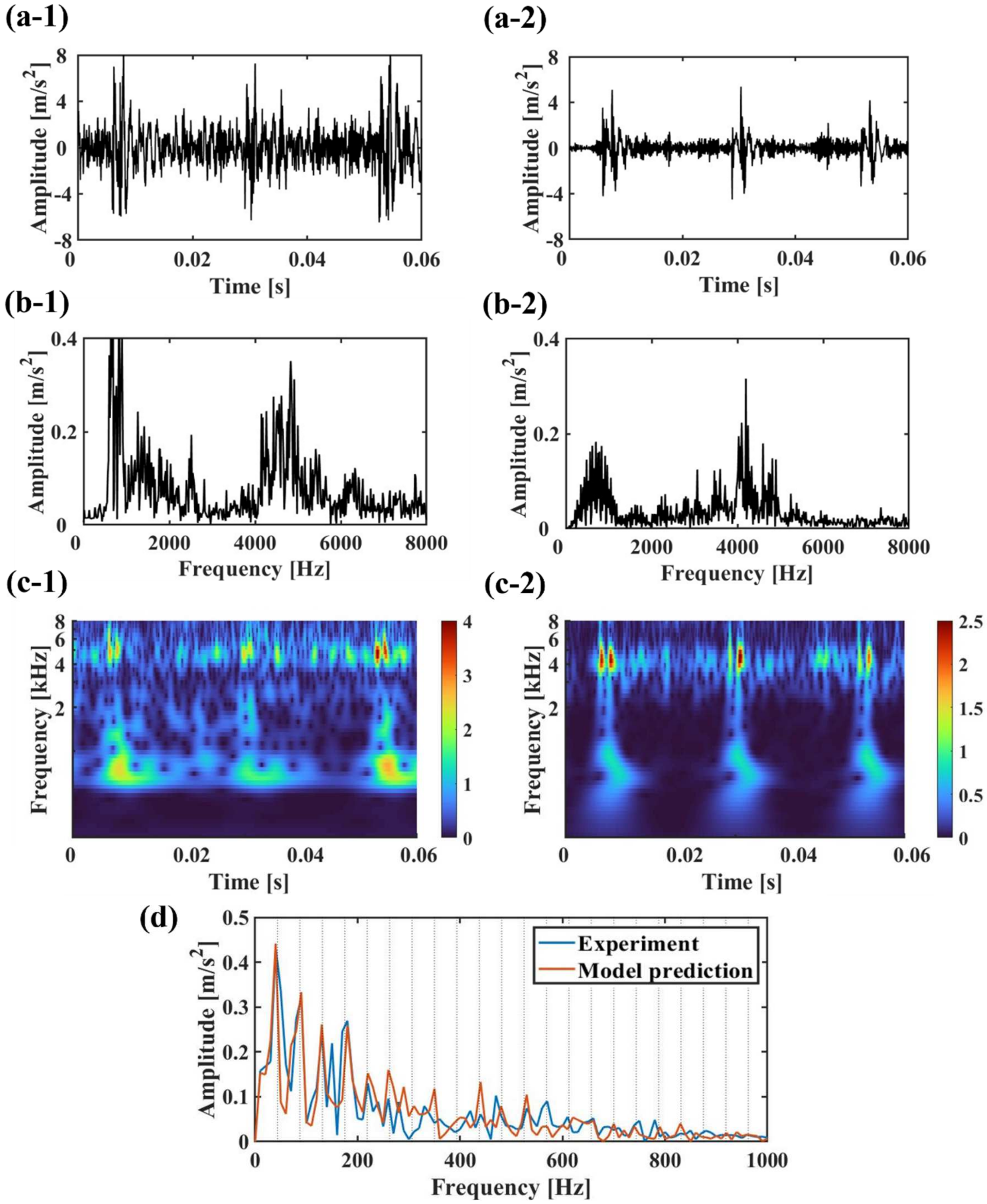

Fig. 6. Vibration signals of 1-mm bearing spalled PMSM at 1,000 RPM operation (left for experiment and right for model prediction) for (a) time domain signal, (b) frequency domain signal, (c) wavelet-based scalogram, and (d) envelope spectrum comparison.

4.4. Vibration response at bearing fault conditions

This subsection analyzes the vibration responses from the proposed model at a bearing fault condition against experimental results. The responses were comprehensively analyzed at a specific spall width and then extended to different spall widths. The bearing defect signal was also processed at each step to facilitate effective and intuitive comparisons.

First, vibration characteristics at a rotating speed of 1,000 RPM with the spall width of 1 mm bearing were compared in terms of each time and frequency domain (Fig. 6(a) and (b)). A time domain signal exhibits periodic peaks at 0.023-second intervals (Fig. 6(a)). This periodicity aligns with the reciprocal of the theoretical BPFO, which is 43.75 Hz for a 1,000 RPM rotation. The envelope spectrum of the vibration response over a duration of 0.1 seconds for each experiment and model are displayed to clearly observe this periodicity (Fig. 6(d)). Gray grids are shown at every multiple of the BPFO frequency. It is evident that both the experimental and model responses align with this periodicity of the BPFO.

Notably, a significant force was generated at each $\frac{1}{\mathrm{BPFO}}$ interval when the REs pass through the spall region, whereas very weak stress was generated at the BPFO frequency when there is no spall in the bearing. The vibration response by this force could be observed in the frequency domain, where a 0.06-second time signal was directly transformed (Fig. 6(b)). This force excited the natural frequencies, especially the first mounting mode and second stator mode. Notably, the first mounting mode was significant, unlike in the case of a healthy bearing condition where it was hardly excited (Fig. 5(b) and (c)). However, the key characteristics, i.e., two distinct forces in a spall fault at the spall entry and exit, was not accessible in the frequency domain. The scalogram was obtained using continuous wavelet transform [44] to enable the simultaneous identification of the system's response and bearing fault characteristics (Fig. 6(c)). The scalogram clearly shows the exact time when the natural frequencies were excited at both the experiment and model prediction. In particular, the second stator mode was not only excited at the spall entry and exit but also consistently excited by electromagnetic sources. Distinguishing between vibrations caused by each source became possible due to the higher magnitude of the stator mode vibration, which coincided with the mounting mode excitation in the spall region.

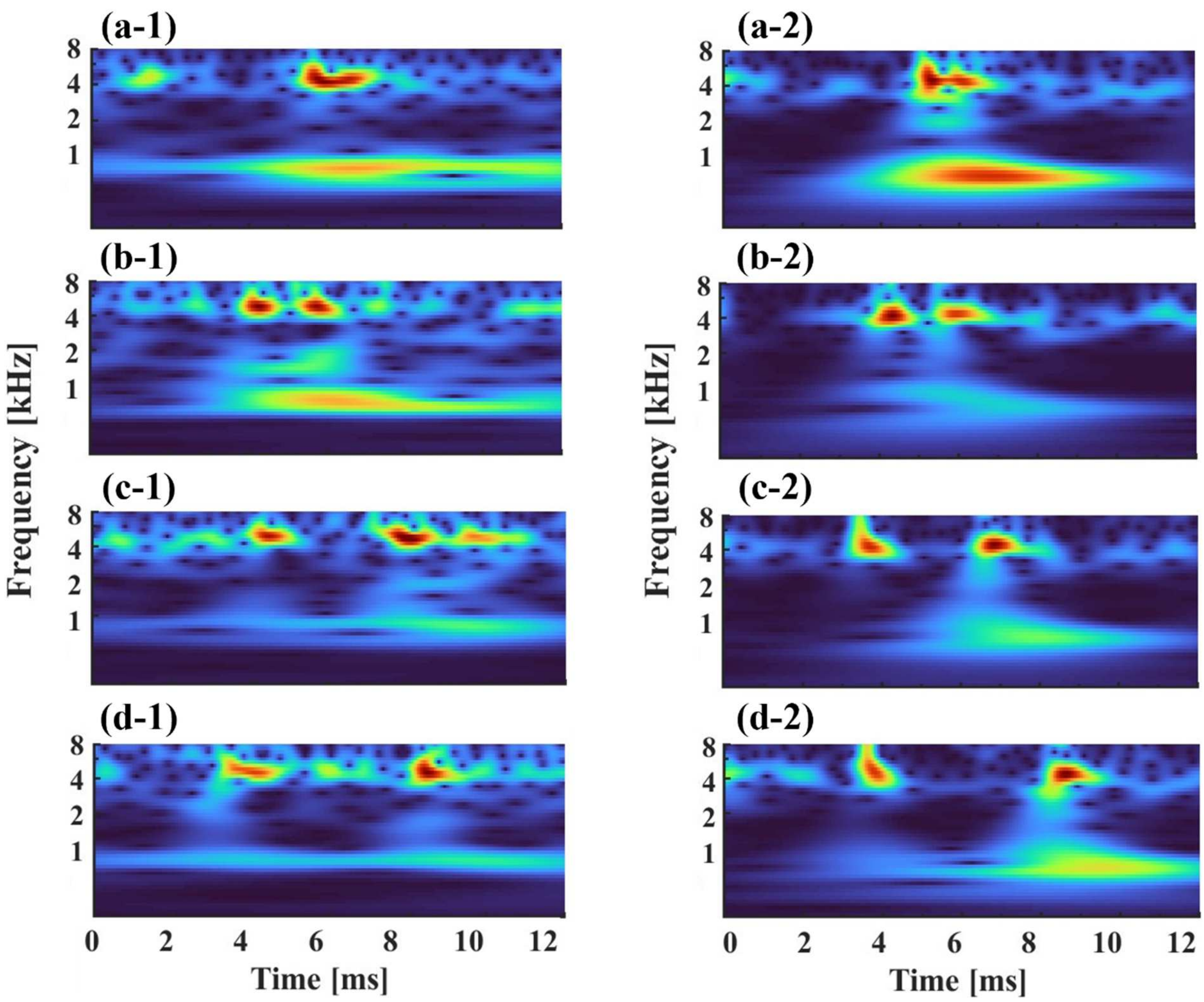

Fig. 7. Scalogram at different spall widths (left for experiment and right for model prediction) of (a) 0.5, (b) 1, (c) 2, and (d) 3 mm.

Signal analysis on the vibration characteristics at a rotating speed of 1,000 RPM with the spall width of 1 mm bearing clearly shows that scalogram would be the most effective to observe the spall fault of bearing. Hence, scalograms was also applied to different spall widths of 0.5-, 1-, 2-, and 3-mm to discern the varying responses across different bearing fault conditions (Fig. 7). One BPFO peak in a time signal, which was comprised of two peaks at the spall entry and exit, was magnified in the scalogram to facilitate a clearer observation of the difference because each BPFO peak would have the same time intervals at all spall widths. It can be observed at both the experiment and model prediction that the time interval between the second stator mode excitation at the spall entry and exit increased proportional to the spall widths. This observation could be explained by the fact that it takes more time for a RE to travel a greater distance between the spall entry location θ_{entry} and exit location θ_{exit}, indicated in Eqs. (9) and (10). Note that the model prediction successfully replicated the responses with a time interval that closely matched the actual measured interval. To quantitatively evaluate the spall size at all cases, the responses of the second stator mode were emphasized by addressing a bandpass filter. Specifically, the time signals were filtered with a passband frequency of 3,500 to 5,500 Hz, near the stator's natural frequency. Then, the vibration magnitude under the root mean square (RMS) was eliminated to reduce the noise including electromagnetic sources, where RMS was calculated using the vibration samples within a single BPFO peak range. This signal was represented in each scalogram and envelope which can demonstrate the intuitive fault characteristics and clear identification of the peak-to-peak intervals, respectively (Fig. 8). The peak-to-peak intervals were pointed in dots at the time axis below the peaks at the envelope. These detected time interval values were substituted into Eqs. (9) and (10) to estimate the spall sizes. Then, the mean and standard deviation values of the estimated sizes were calculated with 15 and 3 samples for the experiment and the model, respectively. Note that only small numbers of samples were used for the model prediction because its responses are much more consistent than the experiment, which has higher uncertainty at REs movement [26]. Fig. 8 clearly shows that the model prediction ensures high accuracy in estimating spall sizes, suggesting that the proposed processing method on a time signal could be effective, which emphasizes the stator mode vibration to detect spall entry and exit. The experimental data tended to overestimate for large spall sizes, despite accurately estimating for small sizes. This discrepancy could be attributed to the presence of the cage structure in the bearing model, which could constrain the movement of the REs. Specifically, the REs in the actual bearing could not descend as deeply as theoretical value due to the cage surrounding REs, resulting in slower impacts on the trailing edge of the spall exit. The standard deviation values were particularly higher in small sizes, likely due to the challenges in accurately detecting very short time intervals. Nevertheless, the mean value remained accurate, indicating the acceptability of the proposed processing methods and equations. It is important to note that both experiment and model yielded similar estimated sizes, affirming the model's accuracy in replicating the actual PMSM behavior, particularly by capturing the two distinct vibrations.

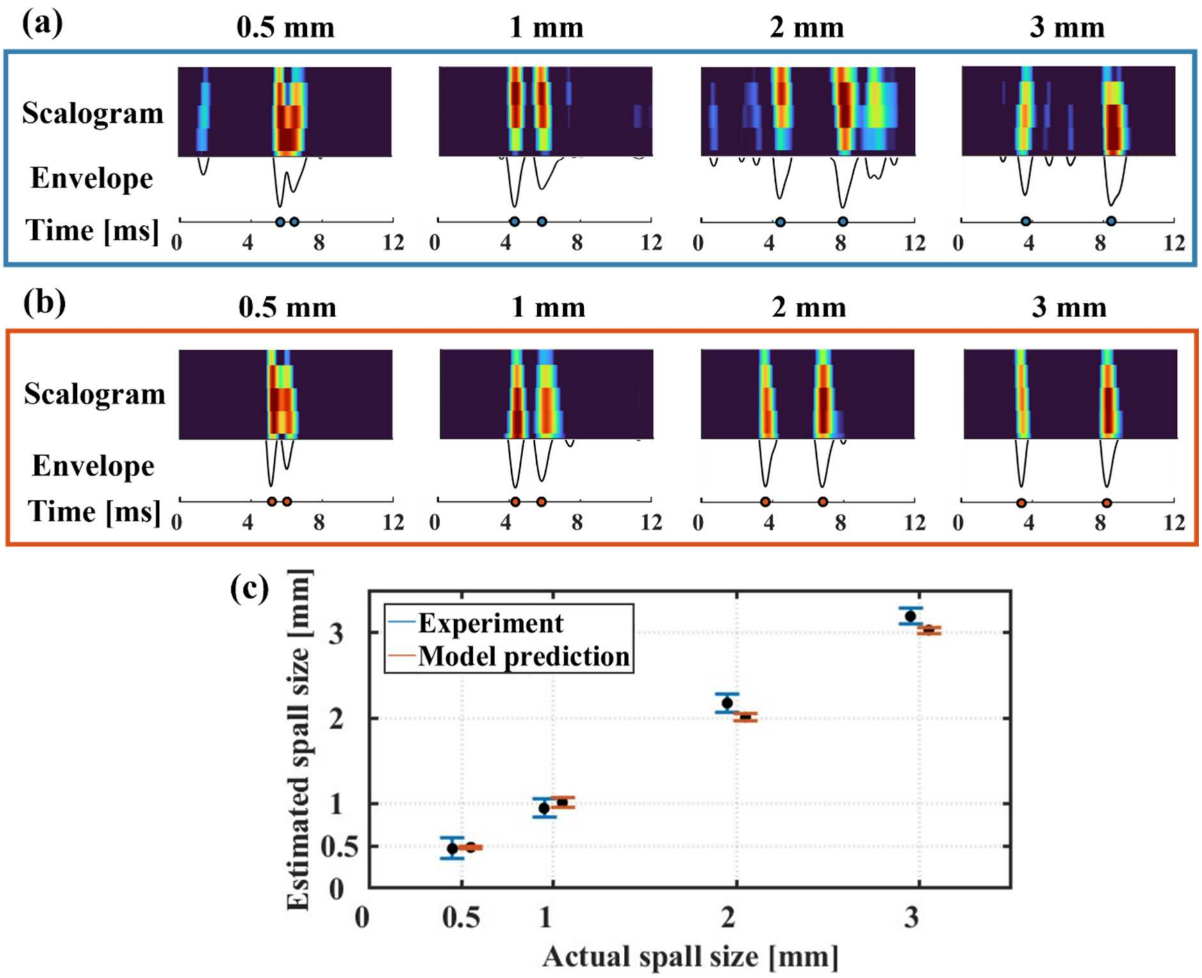

Fig. 8 Processed scalogram and envelope at each spall size for (a) experiment and (b) model prediction, respectively, and (c) the estimated spall size from experiment and model prediction.

4.5. Applications of virtually generated data on fault diagnosis

In order to showcase the application of virtually generated data from the proposed model, a simple data-driven method of fault diagnosis is addressed in this subsection. Specifically, a variational autoencoder (VAE) was employed as a representative data-driven method of fault diagnosis because this method is widely used for fault diagnosis on mechatronics systems [45]. Note that VAE could operate in a latent space, capturing and expressing extracted features through data distribution, which could commonly be used for fault classification and remaining useful life prediction [46].

The architecture of the VAE used in this study is depicted in Fig. 9(a), primarily comprising an encoder and a decoder responsible for compressing and reconstructing high-level features, respectively. The architecture includes a fault size label block where the model is trained to minimize loss, encompassing the combined errors of reconstruction, Kullback-Leibler divergence, and fault class [45]. The model, trained with input vibration data, compresses the data into a two-dimensional latent space upon passing through the encoder. This compression allows data with similar high-level features to be positioned closely together in the latent space, enabling a clear visualization of the fault characteristics within the input data. The model regulates the value in the latent space from 0 to 1 to effectively distribute each fault class data, where 0 and 1 represent the normal state bearing and a 3 mm spall bearing, respectively, with other classes interpolated between these two extremes. The VAE's training utilized solely virtually generated data, incorporating 20 datasets from various locations in the proposed 3D structural model for each bearing spall size.

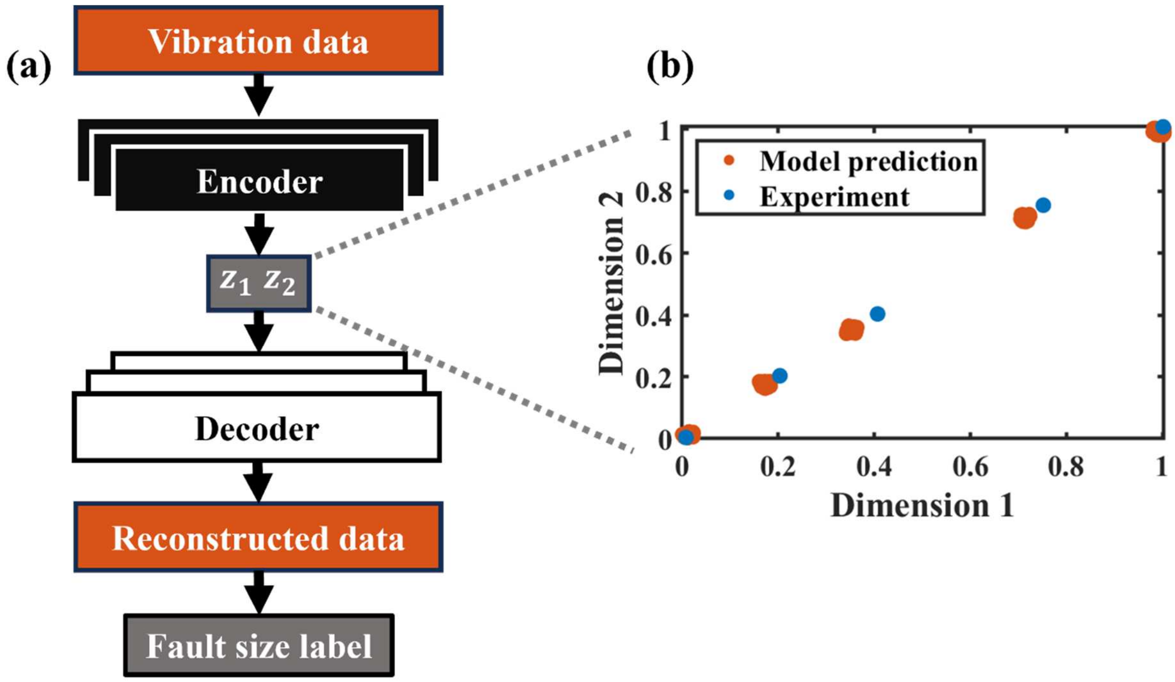

Fig. 9 (a) Architecture of VAE for fault diagnosis and (b) latent distribution of the model predicted vibration data and latent prediction of the experimental vibration data.

Fig. 9(b) illustrates the two-dimensional latent space after the encoder process, showcasing a distinct distribution of 100 datasets based on different spall sizes. Subsequently, the positions within the latent space corresponding to experimental vibration measurements were examined. The outcomes indicate a close alignment between the experimental vibration data at each spall size and similar locations within the virtually generated data. The slight discrepancies exist between the distributed model data and experimental data, which would be originated from hidden dynamics that is not accounted from the model. These discrepancies would be decreased by elaborating the model in modeling perspective. These variances can also potentially be addressed through transfer learning or domain adaptation networks [13, 14], which is beyond the scope of this study. Notably, the proximity of experimental data points to virtually generated data points holds significance in two aspects. First, it validates the fidelity of the generated data by observing similar feature exhibited in the latent space with the experimental data. Second, it lays the groundwork for data-driven fault diagnosis in PMSM bearings, wherein the experimental spall size can be accurately diagnosed by observing its location within the latent space. This approach enables the assessment of experimental spall sizes with data-driven methods without utilizing actually measured fault datasets, demonstrating the effectiveness of generating high-fidelity virtual fault data.

5. Conclusion

This paper proposes a novel method for generating vibration data to study a diagnostic method especially focusing on bearing faults in PMSMs. The proposed modeling method efficiently incorporates multiphysics and dynamic phenomena associated with PMSMs by integrating electromagnetic, REB, and structural models. The model parameters were rigorously estimated through experiments, including torque experiments and modal tests, to faithfully replicate the electromagnetic and structural characteristics of the PMSM of interest. Validation of dynamic responses generated from the proposed model against experimental measurements confirms the reliability of the generation method with the proposed model under both normal and faulty conditions. Furthermore, the accurate estimation of bearing spall sizes is demonstrated considering the dynamic characteristics of PMSMs by processing both experimental and virtual data. The proposed high-fidelity model, capable of generating realistic bearing fault data, contributes significantly to advancing the field of diagnostic and health assessment methodologies for PMSMs. Future work includes expanding the scope of data generation to encompass various potential faults, such as rotor misalignment and inter-turn short circuit. Moreover, health status assessment data-driven algorithm will be developed using these broader fault datasets.

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Author contributions: Hyunseung Lee: Methodology, Formal analysis, Writing - Original Draft. Seho Son: Data curation, Resources. Dayeon Jeong: Validation. Byeong Chan Jeon: Visualization. Ki-Yong Oh: Writing - Review & Editing, Supervision, Project administration. Kyung Ho Sun: Conceptualization, Software.

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